2026–2027 AIDaS Seed Grant Program

Semi-Autonomous AI Research on the Wayne State Grid: Building a Customizable Spine RAG-LLM in One Year, with Medical Trainees Driving the Agents

Strategic areas	Foundational AI (area 1, primary) and AI for Science (area 2), Aim 1: the reliability and cost of agentic systems, studied in agents doing research held to sealed tests, checked code and reproducible records. AI for Education and Workforce Training (area 3; Aim 2): medical trainees doing computational research, with literature answers grounded in verbatim, cited quotations. AI for Health (area 4; Aim 3): a customizable, locally hosted spine RAG-LLM that reads the image through the consortium's detectors and answers with cited quotations, and an advisory level label per C-arm image that the surgeon confirms.
Principal Investigator	Zhizhong Han, PhD, Assistant Professor, Department of Computer Science, College of Engineering, Wayne State University; founding director of the Machine Perception Lab; AIDaS affiliated member. Contact: [DATA NEEDED: office, phone and e-mail].
Co-Principal Investigator	Loren Schwiebert, PhD, Professor, Department of Computer Science, College of Engineering, Wayne State University; AIDaS affiliated member. Contact: [DATA NEEDED: office, phone and e-mail].
Clinical collaborator	Gregory Schwing, MD, PhD, General Surgery, Wayne State University School of Medicine / Detroit Medical Center; leads the Open Spine Consortium. Receives only line C3.
New Investigator status	The PI is a New Investigator: a tenure-track Assistant Professor at Wayne State since January 2021, and an NIH Early Stage Investigator until November 2027, ten years after the PI's PhD of November 2017. The Co-PI is a Professor and not a New Investigator.
Request	$19,987 for 12 months, January 1 to December 31, 2027. No award funds go to an external collaborator: lines B to D buy goods and services, and Aim 3 uses WSU's own C-arm. The PI leads no other AIDaS proposal this cycle.

Abstract. Wrong-level spine surgery is a never event, unpaid by Medicare since 2009 [21] and reported to the Joint Commission 9 times in 2024, 7 of them spinal [22]; paid US malpractice claims for surgical never events reached $1.3 billion over 1990–2010 [23]. It persists because surgeons confirm the level by counting vertebrae on an X-ray from a C-arm, the operating room's mobile fluoroscope, and the count fails in patients with an extra or missing lumbar segment and on images that show too few vertebrae. This project's goal is a customizable, locally hosted retrieval-augmented language model for spine care (a RAG-LLM): a multimodal model on the WSU grid that reads the image through the Open Spine Consortium's X-ray detector and level model, retrieves from the consortium's index of Scopus and ScienceDirect spine literature (text, images and multimedia, under WSU's text and data mining rights) and answers with verbatim, cited quotations; fine-tuned, in its second release, on images paired with clinical charts under the consortium's pending IRB protocol (about 4,000 Detroit Medical Center spine surgery patients; amended as the work requires); compared with OpenEvidence's clinical evidence service through a Claude plug-in [31]; and customizable in that a lab swaps in its own corpus, detectors and records. The consortium's spine program is the case example of what an organized, AI-assisted workflow builds in a short time: an imaging platform (211 catalogued sources, 115 on the grid at 5.3 TB: a census of 21,622 public CTs mined for the anatomy on which counting fails, 39,705 plain radiographs, COBRA's 692 radiotherapy cone-beam cases with 375,618 projections); 802 CTs with every vertebra labeled (CTSpinoPelvic1K, a NeurIPS 2026 benchmark); a physics renderer; and a level model that, never trained on a C-arm image, names the level of 96.4 % of the lateral vertebrae a film-trained detector finds on a public C-arm dataset, while detection on that C-arm stays at 80.5 % (90.6 % at a lower threshold) against the 97 % set in advance, 145 of COBRA's 692 cases among the frames that carry it that far. Aim 1 studies the workflow: an AI agent iterates the scientific method inside the consortium's frameworks (the cluster's job rules, a code gate, reporting protocols, multi-agent coordination on GitHub), a result counting only when a test sealed before the run passes and a person merges it, on tasks with a planted fault submitting a wrong result, we expect, in under a fifth of runs against over half without the checks; from the same records Aim 1 measures what each run costs in tokens, tool calls and context, and Symphony's claim protocol against the worktree and framework orchestrators it competes with, on CooperBench's pairs and SWE-bench. Aim 2 gives the agent to eight medical trainees and measures, task against matched task, what it adds over a coding assistant; their literature questions go to the RAG-LLM on the grid. Aim 3 builds the case example through to the RAG-LLM: the detection gap closed with 372,000 unseen COBRA projections, learned X-ray simulation and a CT-paired cadaver set imaged at WSU, then the multimodal model that reads the image and the chart. Deliverables: Aim 1's tasks, records, benchmark suites and report (MIT license, as WSU's technology-transfer review clears), the RAG-LLM's first release on the public corpus with its graded question set, the cadaver set, an FDA Pre-Submission, and NSF CAREER and NIH R21 proposals.

Keywords. agentic systems; AI for science; reproducibility; coordination protocols; retrieval-augmented generation; multimodal models; high-performance computing; medical education; fluoroscopy; spine surgery

Project narrative

1. Significance and innovation

Surgeons confirm the level of a spine operation by counting vertebrae on an X-ray from a C-arm, the operating room's mobile fluoroscope. The count fails often enough to matter: a 27-hospital Japanese registry recorded wrong-level surgery in 0.12 % of 34,544 patients having degenerative spine surgery [1]; US neurosurgeons reported 418 wrong-level operations in about 1.3 million, one in 3,110, 17 % followed by legal action or a settlement, and about half of respondents report at least one [2,3]. It fails most on a transitional vertebra, part lumbar and part sacral: 12.9 % of 550 surgical patients had one, and their levels were miscounted in clinic in 54 % of them [4]. On 62 hard intraoperative images, surgeons' median naming error was 30.4 % unaided and 2.4 % with LevelCheck [6], which registers the preoperative CT to each image [5] in 17 to 72 s, at the edge of the 20 s to one minute surgeons said they would wait [7]. The hospital's stake is written in rules: wrong-site surgery is a serious reportable event, unpaid by Medicare since January 2009 [21]; the Joint Commission's 2024 review of voluntarily reported sentinel events holds 127 wrong surgeries, 9 wrong-level, 7 of those spinal, 36 % with severe permanent harm [22]; paid US malpractice claims for surgical never events totaled $1.3 billion over 9,744 claims in 1990–2010 [23], 978 English spinal negligence claims over five years £535.5 million [24], and one US insurer's 68 wrong-site claims closed in 2013–2020 averaged $136,453, spine the commonest procedure [25].

The Open Spine Consortium at Wayne State has built the parts of a check that needs no CT, only the C-arm image, and the data platform under it: a card for each of 211 imaging sources, 115 of them on the grid (5.3 TB), every item keyed to a person, nine leakage checks (seven run on the frozen split), and 563 quoted numbers audited against their models' training persons. Its CT census, 21,622 CTs from 21,603 persons, was mined for the anatomy on which counting fails: two automatic labelers chosen to fail differently filled a registry of 6,108 persons from the collections then in hand, 146 of them with a sixth lumbar vertebra (Figure 1d), 103 on an expert read, and 14 with a thirteenth thoracic, and screened 18 further public collections (843 GPU-hours) under a stopping rule, fixed before any download, that turned 29,479 mostly chest CTs away. A labeler's 9.3 % rate of thirteenth rib-bearing vertebrae in 367 chest CTs fell to 1.4 % (5 of 367) once a rib on the named body was required, the check outside the computation that this project's gates make routine. From the platform come CTSpinoPelvic1K, 802 CTs with every vertebra and the sacrum labeled [8], whose benchmark paper is a NeurIPS 2026 poster; 39,705 plain radiographs, 12,709 of which train a film detector that finds 98.8 % (at 0.25) of 161 lumbar vertebrae on 17 spines it never saw; the physics renders and real frames of Aim 3, COBRA's 692 cases among them [29]; and a level model that names 96.4 % of the lateral vertebrae the film detector finds on a public C-arm it never saw, while detection on a C-arm held out of training stays at 80.5 % (90.6 % at a lower threshold), short of the 97 % set in advance (Figure 1f). Closing that gap takes many GPU jobs run, checked and recorded on a shared cluster; the consortium's 2026 trainee program hit the same wall, relying on mentors expert in Python, imaging and cluster computing [9].

Our premise is that the bottleneck has moved from the model to the experiments; that an AI agent can run them if strict frameworks, not the model, decide which results count; that the same frameworks, which record every run, let us measure the agents; and that what such a workflow builds in a year is best shown on one case, this spine program, whose end product is a customizable, locally hosted RAG-LLM: the consortium's detectors reading the image, its indexed literature answering with cited quotations, a multimodal model fine-tuned on images paired with charts (a stated purpose of the pending protocol), compared with OpenEvidence's evidence service through a Claude plug-in [31], with no indexed passage and no patient record leaving the grid. Three aims test it (Figure 1). **Aim 1** has the agent iterate the scientific method inside the consortium's frameworks: Beethoven runs Claude Code on the grid and keeps the grid's profile as checked data and its job rules in writing; beethoven-qa checks the agent's code; beethoven-research holds REFORMS [10] rules for machine-learning science and plans the CLAIM [11] checklist for imaging AI; and Symphony (public, MIT) gives each piece of work to one agent through a claim that is a git branch, which GitHub lets exactly one of two racing agents create, with the board and the release of idle claims as deterministic CI jobs and no server. The methodological contribution measures *framework-bound scientific iteration*: how many wrong but plausible results a research agent submits when the rules are only written, and when checks enforce them, and, from the same records, what its work costs and how the protocol fares against the orchestrators it competes with. Benchmarks score agents' code edits [12], machine-learning engineering [13] or papers [14], and studies of research agents probe their leakage and tampering [15,16]; CooperBench found two agents in separate workspaces, talking only by message, passing 183 of 652 feature pairs against 315 for one (GPT-5), most failures duplicated or incompatible work [27]. We plant the faults an agent inherits from a lab's data and code, count those it passes on, and meter the coordination. **Aim 2** puts the agent in medical trainees' hands through Beethoven (Figure 1b). **Aim 3** builds the case example: it turns the agent on the detection gap, with COBRA's unused projections, a paired CT and C-arm cadaver set imaged at WSU and a differentiable 3-D Gaussian X-ray renderer, as in [17,18], built on the PI's Gaussian-splatting work, and then on the multimodal model that reads the image and the chart.

2. Aims, approach and preliminary work

Aim 1 (Foundational AI, AI for Science): semi-autonomous research under strict frameworks, and the measurement of the agents. The agent is Claude Code at a pinned model version. Semi-autonomous: a person sets each task's question and gate (a pass/fail test, sealed by hash before the first job, that the agent can read but not change) and merges each result; any other step a person takes, blind to faults, is a logged intervention. The agent iterates (Figure 1a): it seals each hypothesis with its criterion, runs jobs under Beethoven's rules, counts a result only from code beethoven-qa passed (contracts searched for counterexamples by CrossHair and Z3) and keeps its REFORMS record, until the gate passes or the budget is spent, then marks its Symphony pull request ready. *1a, framework-bound iteration.* Thirty tasks, 10 for development, come from questions Aim 3 and the consortium have answered. Each of the 20 sealed has a hidden check, a known answer the agent cannot read, and a clean and a faulted copy: once the checks freeze, someone who did not write them plants a fault, a patient in both training and test sets, the wrong label scheme, a threshold tuned on the test set, a typed number, or a run that does not reproduce. Both arms keep the model, Beethoven, the written rules and eight GPU-hours a task, reruns included; one adds the checks (sealed hypotheses, beethoven-qa, split and rerun checks); each runs each copy once (80 runs). *1b, the cost of the agent.* Symphony records what each run cost (a usage trailer on every commit, a usage artifact per Actions run); 1b adds a profiler (agentprof's hooks [28], system-call tracing) that logs every tool call, file read and token to a reproducible trace, and has the agent run, as sealed tasks whose gate is the same tests passing at lower cost, four economies: context compression, fewer tool calls, less code reading (a content-hashed index of the repository and retrieval over it) and profiled prompts; and a standing routine proposes the next experiment from the last record: continuous, autonomous research under the gates. *1c, the protocol against its competitors.* The alternative to a claim protocol is an orchestrator: a script that fans tasks out to agents in git worktrees, merges, reviews adversarially and verifies, as the consortium's own first attempt did; frameworks that make such a script durable (LangGraph's checkpointed fan-out, Temporal's sagas, OpenHands sandboxes, typed outputs); or a shared workspace with optimistic concurrency, which raised a worktree baseline from 63.8 to 82.5 % on Commit0-Lite [26]. 1c runs Symphony, the worktree orchestrator and one framework orchestrator on the

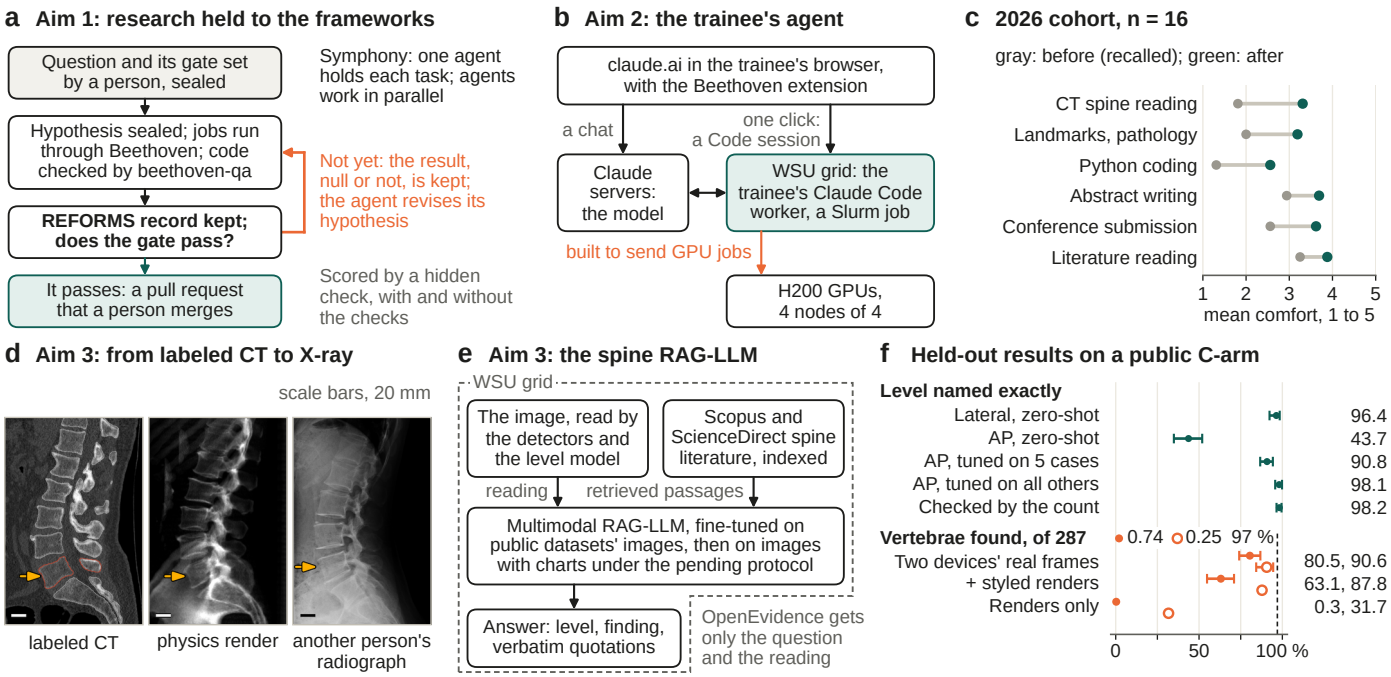


Figure 1. Three aims. (a) The agent iterates until a sealed gate passes. (b) Beethoven starts it on the grid. (c) Mean comfort [9]. (d) A sixth lumbar vertebra (arrow). (e) Inside the dashed line: the grid; answers are graded beside OpenEvidence's. (f) Zero-shot and detection rows never saw C-ArmLSpine; others hold out whole cases. Bars: 95 % case-clustered intervals, where recorded.

same suites in the same Actions environment, coordbench on CooperBench's pairs first [27], then SWE-bench instances [12] and Aim 3's tasks, scoring pass rate, tokens, dollars, wall time, conflicts and duplicated work, and places each system, with gh-aw's work queue, unsorry, Copilot's cloud agent and Claude Code's agent teams, on the twelve-axis taxonomy written for the comparison. Compute is GitHub's free standard runners on public repositories and the grid; inference is line C1's seats, no API key. *1d, the control plane.* A web control panel driven by Claude's own browser (repositories, board, CI, tasks, workers, prompts) showing each run's cost beside the same work on unsorry or Copilot's cloud agent; a wizard storing a lab's Claude tokens as saved environment secrets; Beethoven bridging Actions to the grid, a worker in Actions submitting the Slurm job and posting its results and logs back to the pull request; and the extension's one click brought from Claude Code to the claude.ai Project each trainee works in. *Preliminary work.* Built: Beethoven's grid profile and Claude Code worker (a Slurm job), beethoven-qa's gate on Beethoven's agent and research code, the patient-split check, Symphony (its tests catch all 93 mutated rules; every interleaving of a few workers is checked against an oracle from its spec), coordbench and the taxonomy. The consortium builds the other REFORMS checks, the CLAIM checklist, one-command reruns of gated code, the profiler hooks, the control panel, the wizard and the bridge in kind. *H1:* the checks cut wrong results on faulted copies from over half to under 20 %, with no fewer passes on clean copies, and each accepted number reruns within a sealed tolerance. *H1b:* the four economies together pass the same gates at under half the tokens. *H1c:* the protocol's cost per passing task is at or under each orchestrator's on every suite, with pass rates inside their paired intervals; a loss is reported as one. *Measures:* wrong results submitted, clean copies passed, faults caught by kind and check, reruns, interventions, GPU-hours per accepted result; for 1b and 1c, tokens, tool calls, context and dollars per passing gate, pass rate, conflicts and wall time, paired by task and suite.

Aim 2 (Education and Workforce): medical trainees drive the agent. *Approach.* Eight medical students and residents in two waves of four each receive a grid account, Beethoven and a gated sub-project; the agent runs the experiments inside the frameworks, the trainee owns the question and the result, and a mentor reviews it. Literature questions go to the RAG-LLM's text half, a language model on the grid that quotes verbatim, with citation and DOI, from the consortium's index of spine literature (Scopus, and ScienceDirect's text, images and multimedia) under WSU's text and data mining and AI rights from Elsevier; no passage trains a model or reaches an outside AI service; a misquoted passage fails a gate, and the mentor marks any quotation that does not support the answer. In week 1 each trainee does two matched starter tasks, none of Aim 1's: one with the agent, one with the 2026 setup (Claude as a coding assistant, the trainee submitting the jobs), task and order counterbalanced; the comfort survey runs before and after each wave; an IRB determination is requested in month 1. *Preliminary work* [9]: in 2026, 16 students, seven with no coding and eleven with no imaging experience, worked on ten sub-projects in that setup; all six domains improved on retrospective ratings (Figure 1c), all ten were submitted to a national meeting as abstracts, and the report's own limits, no control group and no objective measure, are what Aim 2 adds. *H2:* with the agent, trainees pass more starter gates without a mentor's edit and reach each task's first passing gate sooner; with eight trainees an estimate with 95 % intervals that sizes the CyberTraining proposal. *Measures:* H2's two, misquoted or unsupported quotations and comfort (secondary), in a preregistered within-trainee crossover.

Aim 3 (AI for Health): the case example, from labeled 3-D structure to 2-D fluoroscopy to a spine RAG-LLM. *Data.* Physics renders of CTSpinoPelvic1K, VerSe and TotalSegmentator CTs (7,045 standard and 1,482 transitional annotated frames) and a device-styled corpus of 142 TotalSegmentator spines; real frames of three devices: C-ArmLSpine (149 public lumbar cases, each a cone-beam CT with AP and lateral frames; truth automatic, two CT labelers agreeing) [30], DeepFluoro (cadaver pelvis fluoroscopy) and COBRA (radiotherapy imaging) [29]. *Model*: a detector finds each body and the sacrum, a level model names each body, a count up from the sacrum checks the names, and the stack abstains where count and shape disagree or the name is unsure. New here, the C-arm detector will learn each body's four corners, as the film detector does, from the CT's corners projected at the device's geometry, exact on renders. *Preliminary results* (Figure 1f). With no C-arm image in training, the level model names 96.4 % of the 617 lateral bodies the film-trained detector finds on C-ArmLSpine (88.0 % of all 676 with agreed truth) but 43.7 % of the 631 AP bodies. Fine-tuned on the AP frames of five cases, it names 90.8 %, and 98.1 % with all other cases, each test case held out. On the 233 frames where the sacrum is found (81.5 % of the 286 that show one), cross-validated models, checked against the count, name 98.2 % of 1,009 bodies, and the stack abstains on

9.9 % of those frames, catching five of the six it would have named wrongly. Detection does not carry: trained on two other devices' real frames, the C-arm detector finds 80.5 % of C-ArmLSpine's vertebrae and 76.8 % of its sacra at 0.74, a cross-device threshold chosen on other devices (90.6 % and 83.9 % at 0.25). *What COBRA adds.* COBRA is 692 radiotherapy cone-beam cases (467 Elekta XVI, 225 Varian OBI), 375,618 projections, each case with its acquisition geometry and a registered planning CT [29]; 145 Elekta cases, 24 frames each, are in any frame bank (110 of them, 2,640 frames, train the detector), and 322 Elekta and all 225 Varian cases train nothing. Those frames carry detection today: Elekta frames and their styled renders alone find 86.8 % (249 of 287 [81.1, 91.8]) of the held-out C-arm's vertebrae at 0.25 on outline truth, cadaver C-arm frames alone 12.9 %, and cutting the Elekta training frames from 2,640 to 240 costs 47.7 points [−53.7, −41.8]. The 372,000 unseen projections, with a CT whose automatic labels project onto every one, are real-image corner labels for the new detector at scale, paired data to fit the simulator's spectrum, scatter and sensor terms, and a second held-out vendor (Varian); the cost is export, storage, a quality look at the Varian bank and GPU hours, an Aim 1 task set. Its limits: a linac's kV imager is not a surgical C-arm; a detector that never saw it finds 6.2 % of its vertebrae (18.5 % at 0.25), the median age of its exported cases is 66 (known for 115 of 145), 7 transitional persons are found and 91 calls unread, and naming on it is weak today (the sacrum defined on 18.2 % of 609 test frames): COBRA trains detection and simulation while the surgical C-arm and the cadaver set test naming. *Closing the gap.* Today's renders hurt: added to real frames, styled renders lower detection from 80.5 to 63.1 % at 0.74 (90.6 to 87.8 % at 0.25), and a classifier tells them from real frames 99 to 100 % of the time. Physics simulation (DeepDRR [19]) with strong domain randomization has matched real-data training for hip landmarks in cadaver X-rays (SyntheX [20]); our device cards fitted to four imagers already reproduce held-out real frames at R^2 0.679 to 0.974. We randomize devices' geometry and spectra likewise, hold out each device in turn, and train 3-D Gaussians to fit each sample in a subset of the labeled CTs, learning only spectrum, scatter and sensor response until that classifier is at chance, so the anatomy and its labels stay exact. *Cadaver set.* Two specimens from WSU's Body Bequest Program are CT-scanned and imaged AP and lateral at lumbar and thoracic positions on WSU's C-arm. A frame grabber records the video output, the set is sealed by hash before any model sees it, and registering each specimen's CT to its frames [5] projects the CT's labels onto every frame. The intended use is an advisory level label on each C-arm image, which the surgeon confirms; two deployment routes have FDA precedents, an app on the C-arm maker's own platform and an external unit reading the C-arm's video output. *The RAG-LLM* (Figure 1e). The detector and level model become tools of a multimodal model hosted on the grid that takes the image, the detectors' boxes and names and the retrieved passages, and answers a clinical question with the level, the finding and verbatim quotations. Its first release is fine-tuned on the public corpus (CTSpinoPelvic1K, the labeled films, COBRA), its second on images paired with operative notes and radiology reports under IRB-26-03-8648, whose stated purposes include a locally hosted multimodal spine LLM, amended as the work requires; a lab swaps in its own corpus, detectors and records. On a held-out question set the mentors write on public images, its answers and OpenEvidence's, through a Claude plug-in [31], are graded side by side for correctness and for quotations that support them; the detectors' reading of an image may go to OpenEvidence with the question, while no indexed passage and no patient record leaves the grid. *H3:* on each held-out C-arm the detector finds 97 % of vertebrae and of sacra with no more false detections than today's at 0.74; the stack names a frame in under 1 s on one GPU (no latency of the stack is on file yet); on every cadaver frame it names the right level or abstains; and on the question set the RAG-LLM is graded correct at least as often as OpenEvidence where the question needs the image, with no unsupported quotation. *Measures:* recall at 0.74 and 0.25, false detections, exact level, abstention, corner error, latency (median, 95th percentile), render-to-real separability, graded correctness and quotation support against OpenEvidence, leave-one-device-out with case-clustered intervals, the cadaver set opened once, models hashed first.

3. Expected outcomes, deliverables and timeline

H1 to H3 and the measures above are written before any run; none is a powered test; H1 and H2 size the CAREER and CyberTraining proposals. Within the award: Aim 1's tasks, records and report, its benchmark suites (efficiency, orchestration, the taxonomy) released through coordbench, the control panel, the wizard and the grid bridge; the cohort's outcomes; the RAG-LLM's first release, on the public corpus, with its question set and grades; the cadaver set, deposited beside CTSpinoPelvic1K as the Body Bequest Program's terms allow, and a transfer and latency manuscript; an FDA Pre-Submission; and the CAREER and R21. The seed buys the researcher's hours, subscriptions, the cadaver set and regulatory review; the trainee program, grid, Beethoven, Symphony, pipelines, the RAG-LLM's first release and faculty time come in kind; the Gaussian renderer is fitted as a feasibility test, its full build left to the R21. *Quarter 1:* missing checks built, tasks and gates sealed, faults planted, the profiler and the COBRA export, recruitment, bookings. *Quarter 2:* Aim 1's runs (months 4–8), 1b's routine, the wizard and the grid bridge, wave 1 (months 4–7), cadaver imaging, transfer and latency results, Pre-Submission. *Quarter 3:* CAREER, 1c's suites, wave 2 (months 8–11), the renderer's fit, the control panel, the RAG-LLM's first fine-tune. *Quarter 4:* R21, the question set graded, final report, symposium. *Risks.* If the checks are not all ready by month 4, Aim 1 tests those built, wave 1 starts on the 2026 setup and the crossover moves to wave 2; agents run attended, a person approving commands blind to faults, until the grid clears unattended runs; if GitHub's or Anthropic's terms bar several Actions workers on one plan, 1c runs on the grid; a delayed specimen leaves the held-out devices as the test; a transfer result under 97 % becomes the R21's first aim.

References. [1] Furuya M, et al. Bone Joint J 2026;108-B(5):673-80. [2] Mody MG, et al. Spine 2008;33(2):194-8. [3] Groff MW, et al. World Neurosurg 2013;79(3-4):585-92. [4] Nakagawa T, et al. Tohoku J Exp Med 2017;242(3):223-8. [5] Otake Y, et al. Phys Med Biol 2012;57(17):5485-508. [6] Manbachi A, et al. Ann Biomed Eng 2018;46(10):1548-57. [7] De Silva T, et al. Spine 2016;41(20):E1249-56. [8] CTSpinoPelvic1K, release 11. doi:10.5281/zenodo.22797014. [9] Schehr A, et al. OpenSpineConsortium. Cureus 2026;18(7):e112661. [10] Kapoor S, et al. Sci Adv 2024;10(18):eadk3452. [11] Tejani AS, et al. Radiol Artif Intell 2024;6(4):e240300. [12] Jimenez CE, et al. SWE-bench. ICLR 2024. [13] Chan JS, et al. MLE-bench. ICLR 2025. [14] Lu C, et al. The AI Scientist. arXiv:2408.06292. [15] Luo Z, et al. arXiv:2509.08713. [16] Atinafu Y, Cohen R. RewardHackingAgents. ICDE Workshops 2026. [17] Cai Y, et al. X-Gaussian. ECCV 2024. [18] Zha R, et al. R^2-Gaussian. NeurIPS 2024. [19] Unberath M, et al. DeepDRR. MICCAI 2018. [20] Gao C, et al. Nat Mach Intell 2023;5:294-308. [21] NQF. Serious Reportable Events, 2011 update; CMS NCD 140.6–140.8 (2009). [22] The Joint Commission. Sentinel Event Data, 2024 Annual Review. [23] Mehtsun WT, et al. Surgery 2013;153(4):465-72. [24] Machin JT, et al. Eur Spine J 2018;27:2693-9. [25] Tan J, et al. Jt Comm J Qual Patient Saf 2023;49(5):265-73. [26] Liu M, et al. Multi-agent collaboration with state management (STORM). arXiv:2605.20563. [27] Khatua A, et al. CooperBench. arXiv:2601.13295. [28] agentprof. github.com/kitaisreal/agentprof, 2026. [29] Thummerer A, et al. COBRA2026. arXiv:2607.20037. [30] C-ArmLSpine. Med Image Anal 2024;97:103283. [31] OpenEvidence, openevidence.com; a Claude Code set-up skill for its API, mcpmarket.com/tools/skills/openevidence-api-setup-auth.

AIDaS funding priority alignment

Target agencies, programs and solicitations

1. **NSF CAREER** (NSF 22-586; CISE, Division of Information and Intelligent Systems), PI Han, built on Aims 1 and 2: framework-bound agents and their benchmarks as the research plan, the trainees as the education plan. Aim 1's agent then runs the 3-D-to-2-D experiments the PI leads in Aim 3, so it serves the PI's own vision program. Deadline July 28, 2027; eligibility is judged at the deadline (an untenured Assistant Professor with no earlier CAREER award). If the PI is not eligible, the same plan goes to IIS's core research under CISE's Future CoRe solicitation (NSF 25-543).
2. **NIH R21 at NIBIB**, built on Aim 3: the Trailblazer Award for New and Early Stage Investigators if NIBIB reissues it (forecast FOR-EB-26-001), or else NIBIB's Exploratory/Developmental Bioengineering Research Grant (PAR-25-321) or the parent R21 (PA-25-304), both open through October 2027. Submitted October 18, 2027 (October 16 is a Saturday), while the PI is still an Early Stage Investigator. Preliminary data: the cadaver set, the held-out-device result and the measured latency; its clinical validation, shaped by FDA's Pre-Submission feedback, draws on the consortium's pending protocol IRB-26-03-8648 (about 4,000 adult DMC spine surgery patients, their CT and records, the RAG-LLM's paired images and charts), where 99 patients give 80 % power to show a Wilson lower bound of 90 % if the true accuracy is 97 %.
3. **NSF CyberTraining** (NSF 23-520), a Pilot of up to $300,000 over two years led by Co-PI Schwiebert: agent-assisted training on advanced computing, sized by Aim 2's estimates. Deadline January 20, 2028; submitted in any case.

Relevance to national science and technology priorities

America's AI Action Plan (July 2025) asks for investment in AI-enabled science and, in its worker-first agenda, for AI skills in the workforce, which Aim 2 builds in Michigan's medical trainees. Executive Order 14363 (November 2025) launches the Genesis Mission, which joins AI agents to national computing, and NSF's Dear Colleague Letter 26-023 invites proposals to existing programs that support it. NIH's Bio Genesis Mission carries it into biomedicine. Aim 1 is that agenda at a university's scale: an agent running science on shared computing, held to verifiable gates. Executive Order 14303, Restoring Gold Standard Science (May 2025), and OSTP's guidance under it (June 2025) ask agencies to promote pre-registration of study protocols and transparent reporting of null results; every gate in this project is pre-registered by hash, and failed gates are reported. NIH's data sharing policy and its Strategic Plan for Data Science 2025–2030 ask for data others can reuse, as CTSpinoPelvic1K is and the cadaver set will be as its terms allow.

Team, expertise and roles

The PI directs the project, writes the FDA Pre-Submission, leads the CAREER and R21 proposals, designs Aim 1 and builds the 3-D-to-2-D model, the Gaussian X-ray renderer and the multimodal RAG-LLM; a computer vision researcher, the PI leads the Machine Perception Lab and publishes on 3-D Gaussian splatting and neural implicit surfaces (CVPR 2025 and 2026, NeurIPS 2024). The PI has advised the consortium's work since September 2026. The Co-PI, a high-performance computing researcher who accelerates scientific codes on GPUs, builds the evaluation harness, plants the faults and keeps the grid measures, and has four such papers with the clinical collaborator (2021–2024). The clinical collaborator leads the Open Spine Consortium, whose dataset, models, Beethoven, Symphony and trainee program [9] this proposal builds on, builds the missing checks and 1b to 1d's tools in kind, and runs the cohort, the IRB determination, the imaging session, the literature index, the RAG-LLM's first release and the pending protocol. Beethoven's repositories are the clinical collaborator's proprietary software, given in kind, with no award funds. Line A's researcher checks Aim 1's runs and curates the cadaver set. Effort: PI 5 %, Co-PI 3 %, in kind.

Timeline for external submission

When	Milestone
January–March 2027 (months 1–3)	Concepts to the NSF IIS and NIBIB program officers; Aim 1's gates sealed; the Pre-Submission written
April 2027 (month 4)	FDA Pre-Submission filed; written feedback due within 70 days
July 28, 2027 (month 7)	NSF CAREER submitted, with Aim 1's early results and Aim 2's first wave
September 2027 (month 9)	The sealed cadaver set opened, once, for the R21
October 18, 2027 (month 10)	NIH R21 submitted, with the cadaver set, the latency and FDA's feedback
December 2027 (month 12)	Project ends; AIDaS final report and symposium
January 20, 2028 (month 13)	NSF CyberTraining Pilot submitted, Co-PI leading
2028	R21 and, if eligible, CAREER (July 26, 2028) resubmitted; NSF Smart Health's next call

The CAREER and R21 meet the expectation of a federal proposal within 12 months of the project's end; the CyberTraining Pilot follows; publications and proposals from the award acknowledge AIDaS.

Budget and justification

Line	Item	Basis	Amount
A	Part-time researcher, hourly	515 h at $15.00: Aim 1 180 h, Aim 2 155 h, Aim 3 180 h	$7,725
A2	Fringe on line A	FICA, 7.65 %	$591
B1	Two cadaver specimens, Body Bequest Program	$1,250 each, estimated	$2,500
B2	Specimen CT, two scans	$300 each, estimated	$600
B3	C-arm session, 0312 Elliman, through DLAR	4 h at $300 an hour, estimated	$1,200
B4	Dosimetry and OEHS review	WSU's fees, estimated	$300
B5	Frame grabber for the C-arm's video output	Magewell USB Capture AIO, retail price	$619
C1	Claude Team Premium seats, in their users' names	6 seats billed annually at $180 each	$1,080
C2	Data management and storage	2 TB of C&IT storage at $400 per TB-year; Zenodo free	$800
C3	Reimbursed Claude subscriptions, 2026	Paid personally by the clinical collaborator, against receipts	$1,500
D	FDA Pre-Submission: regulatory consultant	10 h at $300; FDA charges no fee	$3,000
F	AIDaS symposium poster	36 by 48 in at $6 per sq ft	$72
	Total requested		**$19,987**

- **A, A2.** An hourly graduate student assistant (the call's graduate student support), or else a post-baccalaureate researcher: 515 hours, most in months 4–9 at up to 20 a week, at $15.00, Michigan's minimum wage from January 1, 2027 (MCL 408.934). Temporary employees and students are not benefit-eligible, so A2 is FICA alone: 7.65 % for a hire who is not an enrolled student; hiring a student, at WSU's composite rate of 1.7 %, would buy about 30 more hours. Aim 1's hours check 80 runs against their hidden checks and records, two hours each, and seal the known answers in 20 hours; Aim 2's go to the cohort's accounts and measures; Aim 3's are 100 curating the cadaver set (de-identifying, labeling, sealing by hash) and 80 checking the agent's leave-one-device-out, latency and RAG-LLM runs against their gates. At $18 an hour the total would be $21,650, over the cap unless the hours fall to 429.
- **B1 to B5.** No public dataset we know of pairs a diagnostic CT of the thoracolumbar spine with C-arm frames of the same body: C-ArmLSpine pairs lumbar cone-beam CTs with frames posed by edge registration, and DeepFluoro pairs cadaver CTs with fluoroscopy of the pelvis. The cadaver set adds WSU's own C-arm. The Body Bequest Program's donor authorization permits de-identified images, released publicly on the program's terms. Each C-arm user enrolls in WSU's dosimetry program and files the training form of OEHS, WSU's safety office, both with the director of DLAR 15 business days before the session. The frame grabber records what the C-arm shows, the input a deployed tool would read; at $619 it is far under WSU's $5,000 capital threshold and becomes WSU property.
- **C1 to C3.** Anthropic's Team plan for scientists lets a verified PI add Premium seats at $15 a month, $180 a year billed annually: four for the trainees, reassigned in wave 2, and two for the people who start Aim 1's runs; the frameworks use no API key. All six are bought in January 2027, since the price holds 12 months from sign-up. The WSU grid's compute is free to WSU researchers and given in kind, and Aim 1's benchmark suites run on GitHub's free standard runners for public repositories. Storage holds the render banks, the COBRA export, the literature index, the cadaver set and Aim 1's run records for the award year; what review clears goes to Zenodo at its end. C3 repays the Claude subscriptions the clinical collaborator bought in 2026 to prepare this project, once the collaborator's pending appointment as WSU adjunct faculty begins, since the call bars award funds to external collaborators.
- **D.** FDA charges no fee for a Pre-Submission filed before October 1, 2027, when a fee proposed for fiscal year 2028 would begin. The filing asks FDA whether an advisory level label on each C-arm image, which the surgeon confirms, is the right intended use; whether the held-out-device and cadaver tests suffice as analytical validation; and what clinical study, against what reference standard, the R21 should plan. A mid-level consultant reviews the PI's package in months 1 to 3.
- **F.** Presenting at an AIDaS symposium is an award expectation; the poster is printed at Medical Communications.
- **Not requested.** PI and Co-PI salary, travel, tuition, capital equipment and indirect costs, none assumed.
- **Allowability and dependencies.** The call's allowable costs include graduate student support, computing and data storage: lines A and A2 (with a graduate student assistant), C1 and C2, $10,196 in all, above the call's smallest award of $10,000. C3 is computing bought before the award year, and lines B, D and F are research costs outside those headings, kept unless the AIDaS office disallows them; B2 to B5 depend on B1. Without B the request is $14,768: Aim 3 tests on the held-out devices' frames, and its 100 cadaver hours go to Aim 1 repeats; without B and D, $11,768; without B, D and F, $11,696.

Biosketch: the Principal Investigator

The two-page NSF biosketch of the Principal Investigator goes here in the submitted PDF. Put the file below beside `main.tex` and compile again, and it replaces this page.

`biosketch_han.pdf`

Biosketch: the Co-Principal Investigator

The two-page NSF biosketch of the Co-Principal Investigator goes here in the submitted PDF. Put the file below beside `main.tex` and compile again, and it replaces this page.

`biosketch_schwiebert.pdf`